



Storm Shelter Challenge

My name _____

Class _____

Date _____



How do we build a shelter that stays standing and dry in a storm?

Big storms bring strong wind and heavy rain. People need shelters that stay standing and keep them dry. Can you build one?



My kit tick when you have it

- ☐ LEGO bricks and a base plate
- ☐ Card for a roof
- ☐ A small piece of tissue
- ☐ The same number of bricks for two shelters
- ☐ A desk fan with three speeds
- ☐ Card for a flat roof and a sloped roof
- ☐ A spray bottle of water
- ☐ Two dry pieces of tissue



Stay safe

- Keep small bricks away from little children.
- Keep fingers away from the fan blades.
- An adult switches the fan on and off.
- Keep water far away from the fan and plugs.
- Wipe up spills so nobody slips.

START HERE: BUILD



Build your shelter

1 · DO IT: TICK EACH STEP

- 1 ☐ Start on a base plate.
- 2 ☐ Build four walls. Leave a door.
- 3 ☐ Put a roof on top.
- 4 ☐ Put the tissue inside. It shows if rain gets in.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	Does it stand up by itself?	How many bricks did you use?
My shelter	☐ Yes, strong ☐ Yes, but wobbly ☐ No, it falls	bricks



Draw your shelter

WORDS TO USE

walls roof door strong wobbly shelter bricks

TEST 1



Tall or wide?

? Which shelter stands up to stronger wind?

1 · MY GUESS: CIRCLE ONE



Tall and thin



Low and wide



Both the same

How sure am I?



2 · DO IT: TICK EACH STEP

- 1 ☐ Build a tall, thin shelter.
- 2 ☐ Build a low, wide shelter with the same bricks.
- 3 ☐ Put the fan one ruler away from the shelter.
- 4 ☐ Try low, then medium, then high.
- 5 ☐ Tap the speed that knocked it over.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	Which fan speed knocked it over?
Tall shelter	Low Medium High It never fell
Wide shelter	Low Medium High It never fell



Draw your shelter: Tall shelter



Draw your shelter: Wide shelter

4 · WHAT HAPPENED? CIRCLE ONE



My guess was right



Partly right



Something different happened



Not sure yet

I found out that

because

WORDS TO USE

wind fan tall wide fell strong blew stable

TEST 2



Flat roof or sloped roof?

? Which roof keeps the tissue drier?

1 · MY GUESS: CIRCLE ONE



A flat roof



A sloped roof



Both the same

How sure am I?



2 · DO IT: TICK EACH STEP

- 1 ☐ Put a flat roof on. Put dry tissue inside.

- 2 ☐  Spray the roof 10 times from above.
- 3 ☐  Look at the tissue. Tap what you see.
- 4 ☐  Fold the card into a slope. Use new tissue.
- 5 ☐  Spray 10 times again, then look.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How is the tissue inside?
▬ Flat roof	 Dry  A bit damp  Soaked
▲ Sloped roof	 Dry  A bit damp  Soaked

 Draw your shelter: Flat roof

 Draw your shelter: Sloped roof

4 · WHAT HAPPENED? CIRCLE ONE

- ☒ My guess was right
 ☐ Partly right
 ☐ Something different happened
 ☐ Not sure yet

I found out that _____

because _____

WORDS TO USE

rain roof flat slope dry damp soaked drip

 **Art corner: My storm-proof house**

Draw and colour your dream storm-proof house. Label the strong walls and the roof.

 **Maths corner: Brick count**

Bricks in one wall × number of walls = _____. Which design used fewer bricks?

 **New words**

Shelter: A place that keeps you safe from weather.

Engineer: Someone who designs and builds things to solve problems.

Stable: Hard to knock over.

Slope: A surface that goes up or down at an angle.

★ **What I can do now** colour the stars

I can build a shelter that stands up by itself.



I can test my shelter with wind and rain, fairly.



I can improve my design after testing it.

